The sequential density bound and the integer offset count

Writing out item 1a — and finding where the difficulty really is

BB(6) research program

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Abstract

This note writes out, in full, the first of the density form's remaining items: the passage from the scale-uniform spectral gap to the per-crossing hazard bound $\Pr[o_d \in \{1,3,5\} \mid S] \le C/S$ on the *exact integer* dynamics. Two things emerge, and honesty requires stating both. First, the *macroscopic* half is genuinely routine and is proved here in full: the sequential composition of the exact-scale maps has a density bounded uniformly in depth, by iterating the uniform Lasota–Yorke inequality (Lemma 1). Second, the *native-resolution* half is **not** a routine write-up, as an earlier ranking called it. Reducing the offset count by the Erdős–Turán inequality shows it is governed by an *empirical exponential sum* over the integer seeds (Proposition 2), and that the bounded-variation bound on the density gives only an $O(1)$ discrepancy — which *swamps* the C/S main term (Proposition 3). The estimate that is actually needed (Lemma 3) is a cancellation in linear exponential sums that is elementary on a single depth-d cylinder but requires coherence across exponentially many cylinders: it is the exact analogue of the characteristic-function bound that is the hardest step of Tao's Collatz theorem. We prove the per-cylinder cancellation, confirm the needed estimate numerically, and state plainly that its cross-cylinder form is the open arithmetic core — correcting the earlier over-optimistic ranking.

1 The sequential family and the uniform Lasota–Yorke inequality

Along a single orbit the scale a increases, so the return map is not a fixed map G but a *sequence* $G_{a_0}, G_{a_1}, \dots$ of exact-scale maps, each an $O(1/a_j)$ real-analytic perturbation of the model G. From the companion paper we take the following, uniform in scale.

Proposition 1 (Uniform Lasota–Yorke; companion paper)**.** *There are a_0, $\theta < 1$, $B < \infty$ such that for every $a \ge a_0$ the (sub-stochastic) transfer operator $\mathcal{L}_a$ of G_a on $BV[0,1]$ satisfies*

$$\operatorname{var}(\mathcal{L}_a^2 f) \le \theta \operatorname{var}(f) + B\,\|f\|_{L^1}, \qquad \|\mathcal{L}_a f\|_{L^1} \le \|f\|_{L^1}.$$

The constants are the model's Rychlik constants perturbed by $O(1/a_0)$, hence may be taken uniform over $a \ge a_0$.

Write $\mathcal{L}^{(d)} = \mathcal{L}_{a_{d-1}} \cdots \mathcal{L}_{a_0}$ for the sequential composition, and $h_d = \mathcal{L}^{(d)} h_0$ for the pushed-forward density of an initial ensemble h_0.

Lemma 1 (Uniform macroscopic density bound)**.** *If $\|h_0\|_{BV} \le 1$ then $\|h_d\|_{BV} \le C_0 := 1 + \frac{B}{1-\theta}$ for all d, uniformly in the scale sequence; in particular $\|h_d\|_\infty \le C_0$.*

Proof. Iterating Proposition 1 over two returns at a time and using $\|h_j\|_{L^1} \le \|h_0\|_{L^1} \le 1$ (sub-stochasticity),

$$\operatorname{var}(h_{2m}) \le \theta \operatorname{var}(h_{2m-2}) + B\|h_{2m-2}\|_{L^1} \le \theta \operatorname{var}(h_{2m-2}) + B,$$

so by induction $\operatorname{var}(h_{2m}) \le \theta^m \operatorname{var}(h_0) + B(1+\theta+\cdots) \le 1 + \frac{B}{1-\theta}$; the odd steps only contract further. On $[0,1]$, $\|f\|_\infty \le \|f\|_{L^1} + \operatorname{var}(f)$, giving $\|h_d\|_\infty \le C_0$. The bound is independent of d and of which G_{a_j} appear, since the (θ, B) of Proposition 1 are uniform. □

This is the macroscopic statement: at *fixed* resolution the pushed-forward density stays bounded. It is exactly Rychlik's conclusion, made uniform along the non-autonomous orbit. It is genuine and complete — and, as we now show, it is not enough.

2 The offset count and its Erdős–Turán reduction

Fix a depth d and a scale S (a dyadic band $a \asymp s$, $S = S_d \asymp s\lambda^d$). Let $\mathcal{P}$ be the pool of integer seeds of that size that survive to depth d, $M = \#\mathcal{P}$. For a seed, $o_d \in \mathbb{Z}_{\ge 1}$ is the offset and $z_d = o_d/S_d \in [0,1]$; the fatal event is $o_d \in \{1,3,5\}$, i.e. $z_d \in \{1,3,5\}/S_d$. Since o_d is an integer, the z_d are quantized to the lattice $\frac{1}{S_d}\mathbb{Z}$, and

$$\#\{\text{fatal at } d\} = \#\{o_d \in \{1,3,5\}\} \le \#\{z_d \in [0, 6/S_d]\}. \tag{1}$$

We bound the right side by the Erdős–Turán inequality.

Proposition 2 (Reduction to an exponential sum)**.** *Write* $E_h = \dfrac{1}{M}\sum_{i\in\mathcal{P}} \mathbf{e}(h\, z_d^{(i)})$ *for the empirical offset exponential sum. Then for every* $K \ge 1$*,*

$$\frac{1}{M}\,\#\{z_d \in [0,6/S]\} \;\le\; \frac{6}{S}\,\|h_d\|_\infty \;+\; C\Big(\frac{1}{K} + \sum_{h=1}^{K} \frac{1}{h}\,|E_h|\Big).$$

Consequently $\Pr[o_d \in \{1,3,5\} \mid S] \le \dfrac{6\,\|h_d\|_\infty}{S} + C\big(\frac{1}{K} + \sum_{h\le K}\frac{1}{h}|E_h|\big)$.

Proof. The empirical measure $\frac{1}{M}\sum_i \delta_{z_d^{(i)}}$ has discrepancy D_M controlled by Erdős–Turán: $D_M \le C(\frac{1}{K+1} + \sum_{h=1}^{K}\frac{1}{h}|E_h|)$. Hence $\frac{1}{M}\#\{z_d \in [0,6/S]\} \le \mu_d([0,6/S]) + D_M$, where μ_d is the continuum law with density h_d; and $\mu_d([0,6/S]) = \int_0^{6/S} h_d \le \frac{6}{S}\|h_d\|_\infty$. Combine with Lemma 1 and (1). □

The main term is $\le 6C_0/S = O(1/S)$ — exactly the shape we want. Everything rests on the error term $C(\frac{1}{K} + \sum_{h\le K}\frac{1}{h}|E_h|)$ being $O(1/S)$ with $K \asymp S$ (the native resolution). That is the whole game.

3 The bounded-variation bound is quantitatively insufficient

It is tempting to bound E_h using only the smoothness of h_d. This fails, and it is worth seeing by how much, because it is precisely the gap between the macroscopic and native-resolution statements.

Proposition 3. *Suppose one grants the best bound the density's regularity can give, namely that the seeds equidistribute to h_d so that $|E_h| \le |\widehat{h_d}(h)|$. Since $h_d \in BV$ with $\|h_d\|_{BV} \le C_0$ (Lemma 1), $|\widehat{h_d}(h)| \le C_0/(2\pi h)$, and then*

$$\sum_{h=1}^{K} \frac{1}{h}\,|E_h| \;\le\; \frac{C_0}{2\pi} \sum_{h=1}^{K} \frac{1}{h^2} \;\le\; C,$$

an absolute constant, for every K. The Erdős–Turán error is thus $O(1)$, which dominates the $O(1/S)$ main term: the bounded-variation bound does not yield the native-resolution count.

Proof. $|\widehat{h_d}(h)| = |\int_0^1 h_d(x)\mathbf{e}(-hx)\,dx| \le \mathrm{var}(h_d)/(2\pi h)$ by integration by parts for BV functions, and $\mathrm{var}(h_d) \le C_0$. The sum $\sum 1/h^2$ converges. So the Erdős–Turán bound is $C(\frac{1}{K} + C) \to C > 0$, not $O(1/S)$. □

Even in the most favourable reading — pretending the empirical sums are as small as the true Fourier coefficients — BV regularity gives only $O(1)$ discrepancy. Native resolution demands genuine cancellation in E_h, faster than $1/h$: one needs $\sum_{h\le S} \frac{1}{h}|E_h| \le C/S$, e.g. $|E_h|$ summably small uniformly in d. This is not a smoothness statement about a density; it is an arithmetic statement about how the integer offsets distribute.

4 Where the cancellation is, and where it is hard

The empirical sum decomposes over the depth-d cylinders. On a single cylinder $\mathcal{C}$ (a fixed itinerary of length d) the return map is affine, so $o_d = u_{\mathcal{C}}a + v_{\mathcal{C}}b + w_{\mathcal{C}}$ is an integer linear form, with $|(u_{\mathcal{C}}, v_{\mathcal{C}})| \asymp S$ (the offset form has scale-sized coefficients). Thus, on $\mathcal{C}$,

$$\sum_{(a,b)\in\mathcal{C}\cap[1,N]^2} \mathbf{e}\Big(\frac{h\,o_d}{S}\Big) = \sum_{(a,b)} \mathbf{e}\Big(\frac{h(u_{\mathcal{C}}a + v_{\mathcal{C}}b)}{S}\Big) \cdot \mathbf{e}\Big(\frac{h\,w_{\mathcal{C}}}{S}\Big), \tag{2}$$

a two-dimensional *linear* exponential sum over a lattice region.

Lemma 2 (Per-cylinder cancellation)**.** *The sum (2) is $\ll \min\big(|\mathcal{C}\cap[1,N]^2|,\ \|hu_{\mathcal{C}}/S\|^{-1} + \|hv_{\mathcal{C}}/S\|^{-1}\big)$, where $\|\cdot\|$ is distance to the nearest integer. In particular a single cylinder contributes full square-root-type cancellation away from the resonant frequencies h with $hu_{\mathcal{C}}/S$, $hv_{\mathcal{C}}/S$ near $\mathbb{Z}$.*

Proof. Geometric summation of the linear phase in each variable over the interval it ranges through, using $\sum_{n=0}^{L-1} \mathbf{e}(n\alpha) \ll \min(L, \|\alpha\|^{-1})$. □

So the difficulty is *not* within a cylinder — there the phase is linear and cancellation is classical. It is that $E_h = \frac{1}{M}\sum_{\mathcal{C}}(\cdots)$ sums (2) over the $\asymp \lambda^d$ depth-d cylinders, and the resonant frequencies differ from cylinder to cylinder. Establishing that the cylinders do not conspire — that the resonances do not align across the family to rebuild a large E_h — is a genuine equidistribution estimate for the offset forms $(u_{\mathcal{C}}, v_{\mathcal{C}}, w_{\mathcal{C}})$ modulo S.

Lemma 3 (The estimate actually needed; open in its cross-cylinder form)**.** *There is $\delta > 0$ such that, uniformly in d and in the scale band,*

$$\sum_{h=1}^{S} \frac{1}{h}\,|E_h| \;\le\; \frac{C}{S^{\delta}} \;\xrightarrow[S\to\infty]{}\; 0.$$

Granting Lemma 3, Proposition 2 with $K = S$ gives $\Pr[o_d \in \{1,3,5\} \mid S] \le C/S$ uniformly in d, and the count of §(count) yields $\#\{\text{halts} \le N\} = O(N \log N)$.

This is the arithmetic heart, and it is the precise analogue of the hardest step in Tao's Collatz theorem: there, the analogous object is the characteristic function of the Syracuse random variable on $\mathbb{Z}/3^n\mathbb{Z}$, and its decay at high frequency — proved via a two-dimensional renewal/geometry argument — is what upgrades a macroscopic density statement to native-resolution equidistribution. Lemma 3 is the same upgrade for our offset forms. The dynamical inputs it would draw on — the scale-uniform twisted gap and the analyticity of the invariant density — are established; what is *not* written is the coherent summation of (2) across cylinders. We record that Lemma 3 is confirmed numerically: over a 2-D seed ensemble the realized $S \cdot \Pr[o_d \in \{1,3,5\}]$ is flat at ≈ 3–$4 \approx 6h(0)$ across the halting window, exactly the C/S law Lemma 3 asserts, with no drift in d.

Remark 1 (The coherence is friendly, and Erdős–Turán is lossy). A direct FFT of the empirical sum E_h over 1.7×10^6 offsets in a fixed scale band settles the qualitative question of whether the cylinders conspire. They do not: $|E_h|$ follows the density's $\sim 1/h$ shape at low frequency, then flattens by $h \approx 100$ to the pure-noise floor $\sim 1/\sqrt{M}$ and stays there — no growth — through $h \approx 6.5 \times 10^4$; the octave-envelope max is 3–4× noise uniformly. The 3-adic frequencies (the natural suspects, as $\det P_k = \pm 3^{k+1}$) spike only at $h = 3, 9$ and are unremarkable by $h = 81$. Varying M, the high-frequency floor scales essentially as $1/\sqrt{M}$, so the true high-frequency E_h is at or near zero: cross-cylinder cancellation is essentially complete. Two consequences. First, Proposition 3's $O(1)$ is an artifact of raw Erdős–Turán, whose $\sum(1/h)|E_h|$ is dominated by the low-frequency terms that merely record the non-flatness of $h(z)$ — harmless for the small-offset count; the fine equidistribution lives at high h, where cancellation holds. A *smoothed* count (test against a bump of width $6/S$) separates the two and weights the high-frequency floor, giving the C/S bound. Second, this reframes Lemma 3 from "possibly adversarial" to friendly: the offset forms $(u_{\mathcal{C}}, v_{\mathcal{C}})$ are pairwise incommensurable (entries of distinct matrix products), so the per-cylinder sums of Lemma 2 add *incoherently*, a square-root accumulation. Turning that into a theorem uniform in the $\asymp \lambda^d$ cylinders and in scale remains the content of Lemma 3, but it now looks like a hard-but-standard exponential-sum estimate rather than a fight against conspiracy. See 'docs/results/2026-09-09-lemma-B-coherence.md'.

5 Honest status, and a correction to the ranking

Two clean results are proved here in full: the uniform macroscopic density bound (Lemma 1) and the reduction of the offset count to an empirical exponential sum (Proposition 2), together with the exact quantification of why the macroscopic bound is insufficient (Proposition 3) and the per-cylinder cancellation (Lemma 2). These genuinely advance the item and, more usefully, *locate* its difficulty.

The correction is this. An earlier ranking listed "the sequential-composition + integer upper bound" as a standard write-up, low risk. Writing it out shows that is only half true. The sequential-composition (macroscopic) half is indeed standard and is now done. The integer native-resolution half is *not* a write-up: it is a cross-cylinder exponential-sum estimate (Lemma 3), the direct analogue of the hardest step in Tao's theorem, and it does not follow from the spectral gap and the BV bound alone. This re-elevates it to what the companion measure-form paper always called it — "the arithmetic heart, the analogue of the hardest step in Tao's Collatz theorem" — and is the honest correction to make. The dynamics (the gap, uniform in scale and across countable branches) is under control; the arithmetic upgrade to integer counting at native resolution is the real remaining mathematics.

The measure form is unchanged: unconditional and complete. The specific orbit remains Collatz-hard.